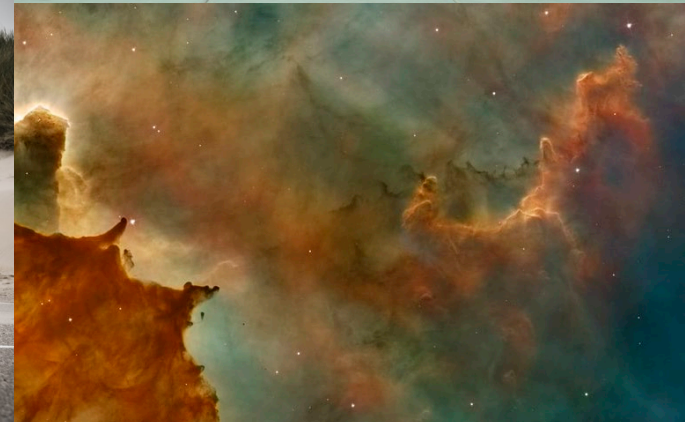
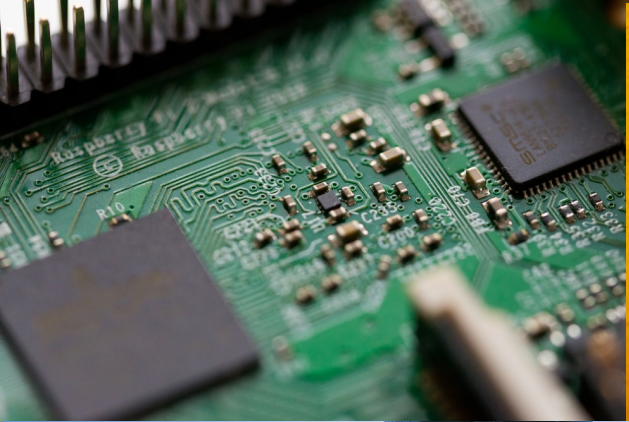
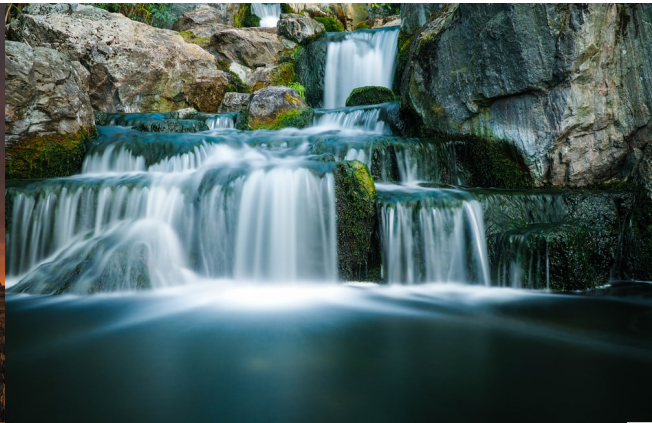




ME 300:
Engineering Thermodynamics

Thermodynamics Applications

Thermodynamics is everywhere!



Example 1: Phase change



liquid $\rightarrow$ vapor, gas

Solid $\rightarrow$ liquid

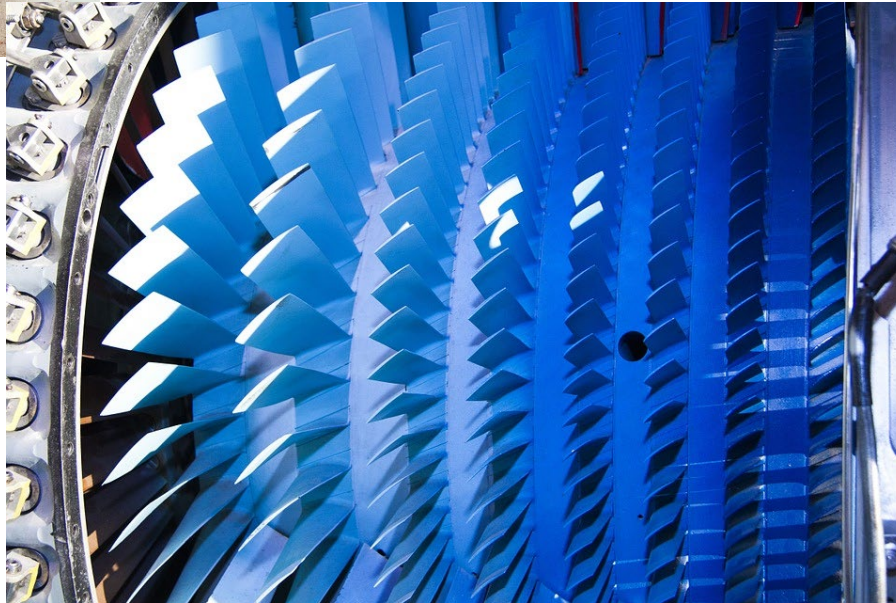
Solid $\rightarrow$ vapor

Example 2: Compression/expansion

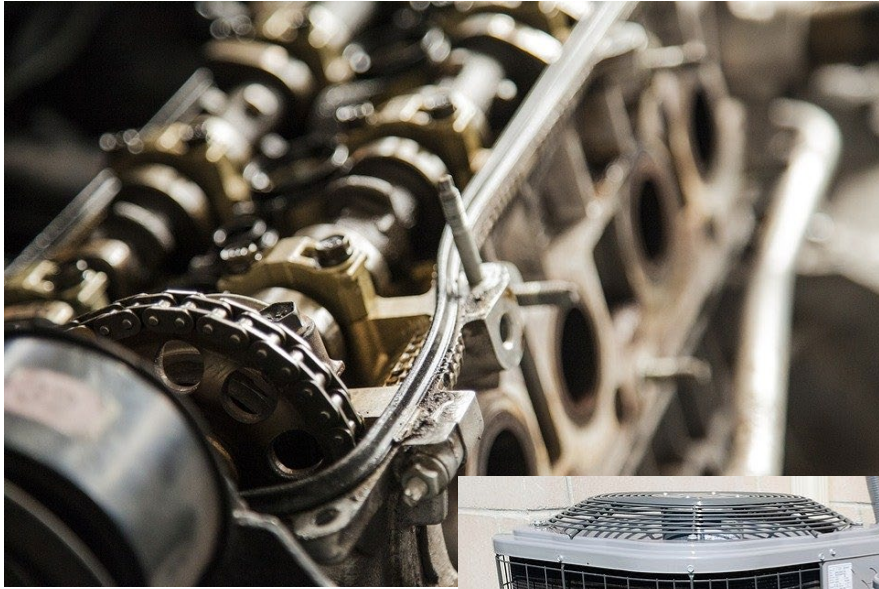


→ energy in by doing work

→ energy out

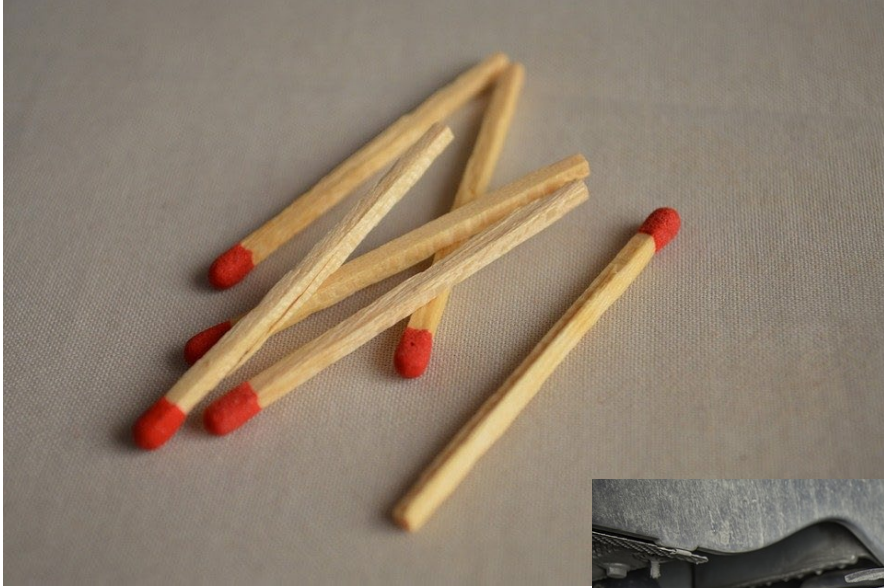


Example 3: Cycles



- fluid comes back to original state
- engine
- heating/cooling

Example 4: Friction loss



friction $\rightarrow$ heat



Classical thermodynamics is largely an empirical science

